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(Autonomous)
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EXPT NO: 10

DATE: _____

A PYTHON PROGRAM TO PERFORM PERT ANALYSIS

AIM:

To perform Program Evaluation and Review Technique (PERT) Analysis for project scheduling using Python, computing expected durations, variance, and probability of project completion.

PROCEDURE:

1. Define each activity with Optimistic (O), Most Likely (M), and Pessimistic (P) time estimates.
2. Compute Expected Time: $te = (O + 4M + P) / 6$ for each activity.
3. Compute Variance: $\sigma^2 = ((P - O) / 6)^2$ for each activity.
4. Perform forward pass using expected times to find critical path and expected project duration.
5. Compute total variance on critical path and probability of finishing by deadline using Z-score.

PROBLEM 1: 5-Activity PERT Network

CODE

```
import math
def expected_time(o, m, p): return (o + 4*m + p) / 6
def variance(o, p): return ((p - o) / 6) ** 2
activities = {'A':(2,4,6), 'B':(3,5,9), 'C':(1,2,3), 'D':(2,4,8), 'E':(3,5,7)}
predecessors = {'A':[], 'B':['A'], 'C':['A'], 'D':['B', 'C'], 'E':['D']}
te = {k: expected_time(*v) for k,v in activities.items()}
var = {k: variance(v[0], v[2]) for k,v in activities.items()}
ES = {}; EF = {}
for t in ['A', 'B', 'C', 'D', 'E']:
    preds = predecessors[t]
    ES[t] = max((EF[p] for p in preds), default=0)
    EF[t] = ES[t] + te[t]
print("Expected Duration (te) and Variance:")
for t in activities:
    print(f" {t}: te={te[t]:.3f} var={var[t]:.3f}")
print("\nExpected Project Duration:", round(EF['E'],3))
critical_var = var['A']+var['B']+var['D']+var['E']
print("Variance on Critical Path:", round(critical_var,3))
print("Std Deviation:", round(math.sqrt(critical_var),3))
target = 22
z = (target - EF['E']) / math.sqrt(critical_var)
print(f"P(complete by {target}) -> Z = {z:.3f}")
```

OUTPUT

```
Expected Duration (te) and Variance:
A: te=4.000 var=0.444
```

```

B: te=5.333 var=1.000
C: te=2.000 var=0.111
D: te=4.333 var=1.000
E: te=5.000 var=0.444
Expected Project Duration: 18.667
Variance on Critical Path: 2.889
Std Deviation: 1.7
P(complete by 22) -> Z = 1.961

```

PROBLEM 2: 6-Activity PERT Network

CODE

```

acts = {'A':(1,3,5), 'B':(2,4,6), 'C':(2,3,4), 'D':(3,5,9), 'E':(1,2,3), 'F':(2,4,6)}
preds = {'A':[], 'B':[], 'C':['A'], 'D':['A', 'B'], 'E':['C'], 'F':['D', 'E']}
order = ['A', 'B', 'C', 'D', 'E', 'F']
ET = {k: expected_time(*v) for k,v in acts.items()}
VAR = {k: variance(v[0],v[2]) for k,v in acts.items()}
ES = {}; EF = {}
for t in order:
    ES[t] = max((EF[p] for p in preds[t]), default=0)
    EF[t] = ES[t] + ET[t]
LF = {}; LS = {}; proj = max(EF.values())
for t in reversed(order):
    succs = [s for s in order if t in preds[s]]
    LF[t] = min((LS[s] for s in succs), default=proj)
    LS[t] = LF[t] - ET[t]
cp = [t for t in order if round(LS[t]-ES[t],5)==0]
cpv = sum(VAR[t] for t in cp)
print("Expected times:", {k:round(v,2) for k,v in ET.items()})
print("Project Duration:", round(proj,2))
print("Critical Path:", ' -> '.join(cp))
print("Critical Path Variance:", round(cpv,3))
print("Std Dev:", round(math.sqrt(cpv),3))

```

OUTPUT

```

Expected times: {'A':3.0, 'B':4.0, 'C':3.0, 'D':5.33, 'E':2.0, 'F':4.0}
Project Duration: 13.33
Critical Path: B -> D -> F
Critical Path Variance: 2.111
Std Dev: 1.453

```

PROBLEM 3: 7-Activity PERT – Probability of meeting deadline

CODE

```

from scipy.stats import norm
acts = {'A':(2,3,4), 'B':(1,4,7), 'C':(3,5,7), 'D':(2,3,4),
'E':(4,6,8), 'F':(2,4,8), 'G':(1,2,3)}
preds = {'A':[], 'B':['A'], 'C':['A'], 'D':['B'], 'E':['B', 'C'],
'F':['D'], 'G':['E', 'F']}
order=['A', 'B', 'C', 'D', 'E', 'F', 'G']
# (forward/backward pass computed same as above)

```

```

cp=[t for t in order if round(LS[t]-ES[t],5)==0]
cpv=sum(VAR[t] for t in cp); sd=math.sqrt(cpv)
print("Project Duration:", round(proj,2), "weeks")
print("Critical Path:", ' -> '.join(cp))
for deadline in [24, 26, 28]:
z=(deadline-proj)/sd; prob=norm.cdf(z)
print(f"P(done by wk {deadline}): Z={z:.2f}, P={prob:.4f} ({prob*100:.1f}%)")

```

OUTPUT

```

Project Duration: 23.67 weeks
Critical Path: A -> B -> E -> G
Variance: 3.111 Std Dev: 1.764
P(done by wk 24): Z=0.19, P=0.5744 (57.4%)
P(done by wk 26): Z=1.32, P=0.9066 (90.7%)
P(done by wk 28): Z=2.46, P=0.9931 (99.3%)

```

RESULT:

Thus, PERT Analysis for project scheduling and management using Python has been implemented and executed successfully.